Corporate financial risk management

Sessions 4-5 · Managing transaction exposure

Learn to identify a foreign-currency cash flow, choose a hedge that offsets its risk, calculate the final receipt or cost, and explain what remains unprotected.

Plan

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1. Why a foreign sale creates currency risk

1.1 Start with the business transaction

- A European firm sells equipment for **GBP 1 million**, payable in three months. Its customer owes a fixed number of pounds, but the firm measures its cash and expenses in euros.
- The contractual pound amount is fixed. The number of euros those pounds will buy is not fixed until conversion, unless the firm hedges.
- If the pound weakens before collection, the same sale yields fewer euros. The customer can pay in full and the seller can still receive less home-currency cash than expected.
- The problem is to manage the **currency value of a payment**, not to change the commercial sale or eliminate every business risk.

> **Transaction exposure:** gains or losses arising from settlement of existing financial obligations whose terms are stated in a foreign currency. This is a cash-flow exposure attached to an existing contract.

1.2 Separate the three FX exposures

Exposure	What can change?	Distinguishing feature
Transaction	Home-currency value of outstanding foreign-currency obligations	Existing contractual receipts or payments
Operating/economic	Future cash flows as FX changes international competitiveness	Future business performance, not just one invoice
Translation/accounting	Reported figures when foreign subsidiaries are consolidated	Currency conversion in consolidation

- Foreign-currency purchases, sales, borrowing, lending, assets and liabilities can create transaction exposure.
- A derivative matching an invoice can protect that transaction. It does not automatically protect future competitiveness or all consolidated accounts.

2. Read the exchange rate before calculating

2.1 A quotation is a price with units

- The **spot market** involves almost immediate purchase or sale of foreign exchange.
- $S_{j/k}$ means the price of **one unit of currency k in currency j**. The denominator is what is priced; the numerator is what pays for it.
- EUR 1.09/GBP means one pound costs 1.09 euros. USD 1.60/EUR means one euro costs 1.60 dollars. These price different currencies and cannot be applied in the same way.

Direct quotation, for a European investor: EUR per foreign unit, such as EUR/USD.
Indirect quotation: foreign units per EUR, such as USD/EUR.
The investor's perspective matters: USD/EUR is indirect for the European investor but direct for a US investor.

Direct quotation = 1 / Indirect quotation
EUR per USD = 1 / (USD per EUR)

Where both quotations describe the same currencies at the same time. Taking a reciprocal reverses the units, not the underlying market.

Worked conversion: USD 100 at USD 1.60/EUR

- **Step 1 - Read:** EUR 1 buys USD 1.60.
- **Step 2 - Reverse:** USD 1 buys EUR 1/1.60 = EUR 0.625.
- **Step 3 - Convert:** USD 100 × EUR 0.625/USD = **EUR 62.50**. Equivalently, USD 100 ÷ (USD 1.60/EUR) = EUR 62.50.
- **Step 4 - Check:** dollars cancel, leaving euros. Multiplication by USD/EUR instead would produce dollars squared per euro, not the wanted currency.

2.2 Tell which currency strengthened

Indirect quotation	EUR value of USD 100	Effect on a EUR-based exporter
USD 1.00/EUR	100/1.00 = EUR 100	Strong dollar: favourable
USD 1.30/EUR	100/1.30 = EUR 76.9231	Intermediate value
USD 1.60/EUR	100/1.60 = EUR 62.50	Weak dollar: unfavourable

- When USD/EUR rises, a euro buys more dollars: **euro strengthens, dollar weakens**. A fixed dollar receipt buys fewer euros.
- When the reciprocal EUR/USD rises, each dollar buys more euros: **dollar strengthens**.
- Do not learn "a higher rate is good". Ask **higher in which quotation, and for a receipt or a payment?**

2.3 Identify the adverse movement

- **Receivable:** foreign currency to collect. The firm will **sell** it. Risk: its home-currency price falls.
- **Payable:** foreign currency to deliver. The firm must **buy** it. Risk: its home-currency price rises.
- General formulas below use a **direct quote: home currency per foreign unit**. Examples explicitly identify when an indirect quote is supplied.

3. Forwards: lock a price and understand the offset

3.1 What the contract obliges the firm to do

- A **forward** contracts today for a future purchase or sale at exchange rate F. In the lecture's setup, no money changes hands at entry; currencies are exchanged at maturity.
- **Long forward:** obligation to buy foreign currency. Matches a payable: the firm needs it for its supplier.
- **Short forward:** obligation to sell foreign currency. Matches a receivable: the firm collects it from its customer.
- A forward is binding even if the later spot price would be better.

Short-forward-hedged receivable: receipt = $Q \times F$
Long-forward-hedged payable: cost = $Q \times F$

Where Q is foreign-currency amount and F is home currency per foreign unit. Amount and maturity must match the commercial cash flow for a full offset.

- First take the number of foreign units; multiply by the agreed price per unit.
- The result is the fixed home-currency amount. For a matched hedge, maturity spot no longer changes this receipt or payment.

3.2 Why a derivative loss can mean the hedge worked

The slide's US importer owes **GBP 100m in one year**. F = **USD 1.50/GBP**.

Long-forward gain/loss relative to spot = $Q \times (S_T - F)$
Short-forward gain/loss relative to spot = $Q \times (F - S_T)$

Where S_T is maturity spot, in the same units as F. Positive is a derivative gain; negative is a loss. Subtract the prices first, then multiply by the foreign notional.

Calculation at maturity	Spot USD 1.80/GBP	Spot USD 1.20/GBP
Buy GBP 100m at spot	Cost USD 180m	Cost USD 120m
Long-forward gain/loss	100m × (1.80 - 1.50) = +USD 30m	100m × (1.20 - 1.50) = -USD 30m
Spot cost minus derivative gain	180m - 30m = **USD 150m**	120m - (-30m) = **USD 150m**

- Stronger GBP makes the commercial purchase USD 30m more expensive; the derivative gains USD 30m and offsets it.
- Weaker GBP makes the purchase USD 30m cheaper; the derivative loses USD 30m and offsets that benefit.
- This decomposition explains the locked rate. With physical delivery, pay USD 150m for GBP 100m under the forward. **Do not count that physical payment and then add the payoff again.**
- A flat hedged-payoff line means no FX gain/loss relative to the locked benchmark. It does not mean the supplier costs zero.

3.3 Hedging versus speculation

- A long forward gains above F; a short gains below F. Both cross zero at F.
- The slide's USD/Swiss-franc diagram uses F = 1.0815. At spot 1.10, the long earns **0.0185 USD per franc**; at 1.05 it loses **0.0315**. The short has the opposite result.
- Without an offsetting commercial cash flow, this is **speculation**. With a matched exposure, judge the derivative and the business payment together.

3.4 Forward premium or discount

A forward price can differ from today's spot price. The lecture expresses that difference as an annualised percentage. It is a **quoted-price comparison**, not an option premium or realised hedge profit.

$$f_{N,j} = [(F_N - S) / S] \times (360 / N)$$

Where:

- S is spot in **USD per unit of currency j**, American terms.
- F_N is the N-day forward quote in the **same units**.
- N is days to maturity; the convention here is a 360-day year.
- f is an annualised decimal; multiply by 100 for a percentage.

Worked example: EUR in the slide's exchange-rate table

- **Step 1 - Match inputs:** S = USD 1.1235/EUR; six-month F = USD 1.1299/EUR; N = 180 days.
- **Step 2 - Difference:** 1.1299 - 1.1235 = **USD 0.0064/EUR**. EUR is priced higher forward than spot.
- **Step 3 - Relative difference:** 0.0064/1.1235 = **0.00569648**, or 0.569648% over six months.
- **Step 4 - Annualise:** 0.00569648 × (360/180) = **0.01139297**.
- **Step 5 - Interpret:** ×100 gives a **1.1393% annualised forward premium on EUR** in this USD/EUR quotation.
- F > S means currency j has a premium in this quote; F < S means a discount.
- Reversing the quote changes the currency whose premium you describe. Do not invert just one rate or label this result a premium on USD.
- Annualisation scales a period difference; it does not turn a six-month contract into a one-year contract.
- The lecture's three-month yen answer is **0.460% annualised**. Printed yen quotes are rounded and do not reproduce it exactly; preserve precision when an exercise provides it.

3.5 Certainty has a trade-off

- A forward removes conversion-price uncertainty on its matched amount. It does not guarantee a better outcome than staying unhedged.
- The firm gives up favourable FX movements in exchange for protection against adverse ones.
- Locking the price on GBP 1m does not ensure the company will actually sell enough goods to collect GBP 1m.
- An option replaces the obligation with a **choice**. That changes the protection from a fixed outcome to a floor or ceiling.

4. Options: insure the downside and retain upside

4.1 Right, strike and premium

- A **call** gives its buyer the right to buy foreign currency at strike K. An importer buys a call to protect against a higher purchase price.
- A **put** gives its buyer the right to sell at K. An exporter buys a put to protect against a lower selling price.
- The buyer can let the option expire when spot offers a better price. The **premium** buys this flexibility and is paid whether the option is exercised or not.
- These examples concern bought options: the firm purchases protection, rather than selling an option to someone else.

4.2 Exporter: a put sets a receipt floor

- The exporter collects Q foreign units. If spot $S_T < K$, exercise and sell at K. If $S_T > K$, sell at spot instead. At equality the gross receipt is the same.

$$\text{Put-hedged net receipt} = Q \times \max(S_T, K) - P_T$$
$$\text{Minimum net receipt} = Q \times K - P_T$$

Where Q is foreign amount, S_T and K are home currency per foreign unit, and P_T is **total home-currency premium at the comparison date**.

- **First:** choose the higher selling price, spot or strike.
- **Then:** multiply by the amount for gross receipts.
- **Finally:** deduct premium. The net floor is lower than the gross strike receipt: insurance is not free.

4.3 Importer: a call sets a cost ceiling

- The importer must buy Q units. If $S_T > K$, exercise and buy at K; if $S_T < K$, buy at cheaper spot instead.

$$\text{Call-hedged total cost} = Q \times \min(S_T, K) + P_T$$
$$\text{Maximum total cost} = Q \times K + P_T$$

The variables use the same units as above.

- **First:** choose the lower buying price.
- **Then:** multiply by Q for the currency purchase cost.
- **Finally:** add premium. A bought option adds to importer cost, never reduces it.

4.4 Compare money at the same date

- If premium is paid today but cash flows are compared at maturity, follow the exercise's financing/opportunity-cost convention.
- With simple annual rate r and horizon t years: $\mathbf{P_T = P_0 \times (1 + r \times t)}$. P_0 is today's total premium. Compute the period rate r×t, then multiply today's cost by one plus that rate.
- Aidan and Boeing importer use this adjustment. Boeing exporter and recap deduct their stated premiums directly; no extra financing rate is supplied there.
- **Strike decides exercise. Strategy break-even compares net outcomes** and therefore depends on premium and forward price too. The examples derive these separately.

5. Aidan: protect a GBP receivable in EUR

5.1 Translate the sale into an exposure

- European exporter Aidan sells equipment to Regency for **GBP 1,000,000**. Sale in March, collection in June: three months.
- The sale is large relative to Aidan's business; it has no other current foreign customers. A weak pound can materially reduce its euro receipt.
- Aidan receives pounds and wants euros: **sell GBP forward or buy a GBP put**.

Input	Value	Function
Receivable Q	GBP 1,000,000	June amount to convert
Spot S_0	EUR 1.09/GBP	Today's value for pricing the premium
Forward F	EUR 1.076/GBP	Fixed June selling price
Put strike K	EUR 1.07/GBP	Optional June selling price
Premium	1.5%	Percent of spot-valued notional
Annual cost of capital	12%	Carry option cost to June
Forecast spot	EUR 1.085/GBP	Management's view, not the strike

5.2 The forward fixes receipts

- **Step 1:** sell GBP 1m forward for June, matching the receivable's amount and date.
- **Step 2:** 1,000,000 × 1.076 = **EUR 1,076,000**.
- **Step 3:** interpret. Spot can end at 1.00 or 1.20; the pounds still sell at 1.076. Aidan has certainty but no benefit from a stronger pound.

5.3 Price the put before calculating its payoff

- **Step 1 - Value the notional:** GBP 1m × EUR 1.09/GBP = EUR 1,090,000 today.
- **Step 2 - Apply 1.5%:** 1,090,000 × 0.015 = **EUR 16,350 paid today**. This is not EUR 0.015 per pound.
- **Step 3 - Period rate:** 12% × 3/12 = **3%**.
- **Step 4 - Carry to June:** 16,350 × 1.03 = **EUR 16,840.50**.
- **Step 5 - Per-pound maturity cost:** 16,840.50/1m = **EUR 0.0168405/GBP**.

Net June receipt = 1,000,000 × max(S_T, 1.07) - 16,840.50

Here S_T is EUR/GBP at collection. Choose the higher selling price, convert the GBP 1m, then deduct the maturity premium.

- **Weak pound, spot 1.00:** exercise. Gross = 1m × 1.07 = EUR 1,070,000; net = 1,070,000 - 16,840.50 = **EUR 1,053,159.50**.
- **Strong pound, spot 1.10:** do not exercise. Sell at spot for EUR 1,100,000; deduct the same premium: **EUR 1,083,159.50**.
- The option protects downside while retaining upside. It does not make the premium disappear in the favourable branch.

5.4 Floor and break-even are different calculations

The floor is the worst net receipt under the put.

- Below 1.07, exercise guarantees EUR 1,070,000 gross. Deduct 16,840.50 for a **EUR 1,053,159.50 net floor**.
- The forward guarantees EUR 1,076,000, higher by **EUR 22,840.50**. The option's advantage is possible upside, not a better guaranteed minimum.

Break-even asks when that upside makes the put equal to the forward.

- On the non-exercise branch, net receipt = 1,000,000 × S_{BE} - 16,840.50.
- Set **1,000,000 × S_{BE} - 16,840.50 = 1,076,000**.
- Add premium to both sides, then divide by GBP 1m: **S_{BE} = EUR 1.0928405/GBP**.
- The candidate is above strike 1.07, so this branch is valid. Below strike, the flat option floor never equals the forward receipt.

Spot (EUR/GBP)	No hedge (EUR)	Forward (EUR)	Put net (EUR)
1.00	1,000,000	1,076,000	1,053,159.50
1.07	1,070,000	1,076,000	1,053,159.50
1.10	1,100,000	1,076,000	1,083,159.50
1.20	1,200,000	1,076,000	1,183,159.50

- **Exercise below 1.07; forecast 1.085; option beats forward above 1.0928405.** A forecast is a view, not a contract term.
- At forecast 1.085, put net = 1,085,000 - 16,840.50 = **EUR 1,068,159.50**, below the forward. Expecting appreciation does not by itself prove the put gives a better net outcome.

5.5 Understand the money-market line in the source chart

This alternative replaces future uncertain conversion with conversion **today**.

- **Step 1 - Borrow pounds now:** GBP borrowing rate is 10% annually, or 2.5% for three months. Borrow 1m/1.025 = **GBP 975,609.76** against June collection.
- **Step 2 - Convert now:** 975,609.76 × 1.09 = **EUR 1,063,414.63**. Future FX movements no longer change these euros.
- **Step 3 - Repay in June:** the customer's GBP 1m pays the GBP loan plus interest exactly. Do not count it again as euro revenue.
- **Step 4 - Carry euros under the chosen assumption:** the chart uses Aidan's 12% annual opportunity return, giving a three-month factor 1.03. Result = 1,063,414.63 × 1.03 = **EUR 1,095,317.07**.
- At the stated 6% annual **deposit rate**, the factor would be 1.015 and proceeds **EUR 1,079,365.85**. These are different carrying assumptions. The opportunity-value line is not the cash earned by a 6% deposit.

Aidan's final choice combines risk tolerance, its FX view and confidence in that view. Boeing applies the same logic first to a receivable, then to a separate payable.

6. Boeing exporter: a forward versus a put

6.1 Identify the contract

- Boeing sells a landing gear system to British Airways for **GBP 10m**, collectible in one year. Home currency is USD; a weak pound reduces dollars received.
- Forward F = **USD 1.46/GBP**. Put K = **USD 1.46/GBP**. Premium = **USD 0.02/GBP**.
- The slides deduct the premium directly in this comparison. No extra financing adjustment is supplied.

6.2 Build each outcome

Forward: fixed selling price

- Sell GBP 10m forward. Receipt = 10,000,000 × 1.46 = **USD 14,600,000**.

Put: floor plus upside, less protection cost

- Total premium = 10,000,000 × 0.02 = **USD 200,000**.
- Below 1.46, exercise: gross receipt USD 14.6m; subtract USD 0.2m: **USD 14.4m net floor**.
- Above 1.46, sell at spot and deduct USD 0.2m. An extra USD 0.01/GBP raises receipts by 10m × 0.01 = USD 100,000.

Forward receipt = USD 14,600,000
Put net receipt = 10,000,000 × max(S_T, 1.46) - 200,000

Here S_T is USD/GBP at collection. Select the selling price; multiply by GBP 10m; deduct the whole premium.

6.3 Derive the break-even

- The put floor is below the forward receipt. Equality must occur on the rising branch above strike.
- Set **10,000,000 × S_{BE} - 200,000 = 14,600,000**.
- Add premium and divide: S_{BE} = 14,800,000/10,000,000 = **USD 1.48/GBP**.
- At **1.46**, spot equals strike but the put still yields USD 0.2m less than the forward.
- At **1.48**, the extra USD 0.02/GBP obtained at spot pays for that premium relative to the forward.

Spot (USD/GBP)	No hedge (USD m)	Forward (USD m)	Put net (USD m)
1.40	14.00	14.60	14.40
1.46	14.60	14.60	14.40
1.48	14.80	14.60	14.60
1.55	15.50	14.60	15.30

- The source graph shows **total dollar proceeds**, not standalone option payoff: forward flat at USD 14.6m; put flat at USD 14.4m until 1.46, then rising and crossing at 1.48.
- For the higher guaranteed receipt, favour the forward. To retain gains from a sufficiently stronger pound, the put can fit, with a lower guaranteed net floor.
- The next case is a different commercial transaction: do not carry this GBP 10m receivable into the importer calculation.

7. Boeing importer: a forward versus a call

7.1 Reverse the exposure before changing instruments

- Boeing imports a Rolls-Royce engine and owes **GBP 5m in one year**. A stronger pound raises its USD purchase cost.
- Boeing must **buy pounds**: long GBP forward or bought GBP call, not a put.
- Spot USD 1.80/GBP; F = **1.75**; K = **1.80**; premium **USD 0.018/GBP**; US annual rate **6%**. UK 6.5% is also in the source table but is not needed for these forward/option calculations.

7.2 Separate purchase cost and insurance cost

- **Forward:** buy GBP 5m at 1.75. Cost = 5,000,000 × 1.75 = **USD 8,750,000**.
- **Premium today:** 5,000,000 × 0.018 = **USD 90,000**.
- **Premium at one year:** 90,000 × 1.06 = **USD 95,400**, or **USD 0.01908/GBP**.
- Compare maturity cash flows: add 95,400 to the currency cost, not today's 90,000.

Call total cost = 5,000,000 × $\min(S_T, 1.80)$ + 95,400
Maximum cost = 5,000,000 × 1.80 + 95,400 = USD 9,095,400

S_T is USD/GBP in one year. Choose the cheaper buying price; convert GBP 5m; add the maturity premium.

Two fully worked scenarios

- **Spot 1.65:** spot is cheaper than strike, so do not exercise. Buy GBP 5m for 5m × 1.65 = USD 8.25m. Add 95,400: **USD 8,345,400**, cheaper than forward by **USD 404,600**.
- **Spot 1.90:** exercise to buy at 1.80 rather than 1.90. Purchase USD 9m + premium 95,400 = **USD 9,095,400**. The call limits the bill, but the forward costs less in this scenario.

7.3 Derive break-even on the valid branch

- Call ceiling exceeds forward cost. Equality can only occur on the lower-spot, non-exercise branch.
- Set **5,000,000 × S_{BE} + 95,400 = 8,750,000**.
- Subtract premium and divide: S_{BE} = 8,654,600/5m = **USD 1.73092/GBP**.
- This is below 1.80, confirming the branch is valid. Call costs less below 1.73092; forward costs less above it.

Spot (USD/GBP)	Forward cost (USD)	Call total cost (USD)
1.65	8,750,000	8,345,400
1.73092	8,750,000	8,750,000
1.80 or higher	8,750,000	9,095,400

- The call total-cost line rises until strike 1.80, then is flat at the ceiling. The forward line is flat at USD 8.75m throughout.
- **Exercise boundary 1.80; strategy break-even 1.73092.** These answer different questions.
- So far, hedge and commercial amounts matched. The next case shows why that condition cannot be assumed for forecast sales.

8. Over-hedging: the sale shrinks, the contract does not

8.1 A budget is not a certain receipt

- **T0 - Budget:** EUR-based exporter expects USD 100 revenue and EUR 70 costs. Budget quote is **USD 1.30/EUR**, indirect.
- Exact expected revenue = 100/1.30 = EUR 76.9231. Margin = 76.9231 - 70 = **EUR 6.9231**.
- The slide rounds the reciprocal to EUR 0.77/USD, hence EUR 77 revenue and EUR 7 margin. Keep full decimals until the final answer.
- If sales stay USD 100, spot 1.60 gives margin 100/1.60 - 70 = **EUR -7.50**; spot 1.00 gives **EUR 30**. These are price effects assuming the same quantity sold.

Over-hedging: derivative notional exceeds actual exposure. The excess derivative has no matching commercial cash flow and becomes a speculative position.

8.2 Follow every cash flow when sales fall

- At T0, sell **USD 100 forward** at USD 1.30/EUR.
- **T1 - Invoicing:** actual sales are only **USD 50**. The contract still requires USD 100: USD 50 is covered by the sale, USD 50 is uncovered.
- **T2 - Collection:** spot is **USD 1.00/EUR**. The dollar strengthened, but the contract amount has not shrunk.

Collection step	Calculation	EUR cash flow
Sell customer dollars under forward	USD 50/1.30	+38.4615
Buy missing dollars to meet remaining obligation	USD 50/1.00	-50.0000
Deliver those bought dollars under forward	USD 50/1.30	+38.4615
Total currency receipt	38.4615 - 50 + 38.4615	**26.9231**
Subtract operating costs	26.9231 - 70	**Margin -43.0769**

- Uncovered USD 50 costs EUR 50 to acquire but brings EUR 38.4615 through the forward. **Excess-hedge loss = EUR 11.5385**.
- Genuine sale receipt minus excess loss minus costs = 38.4615 - 11.5385 - 70 = **EUR -43.0769**, approximately the slide's -43.
- The hedge still matched the actual sale on USD 50. The problem is the other USD 50: no sale offsets that derivative obligation.

8.3 Explain what the total loss contains

- Budget margin 6.9231 to final margin -43.0769 is **EUR 50 deterioration**.
- **Revenue shortfall at locked rate:** USD 50/1.30 = EUR 38.4615. **Excess-hedge loss:** EUR 11.5385. Together: EUR 50.
- Do not call all EUR 50 a derivative loss: quantity risk and excess-hedge loss are distinct.
- Design questions: **when** to hedge (budget or invoicing), and **how much** (100% or less). Fixing an FX price does not fix sales volume.

9. Recap: budget, invoicing and collection

9.1 Put the time line and units in order

- EUR-based exporter: expected **USD 100 at T0**, invoice **USD 100 at T1**, collection at **T2**. Unlike over-hedging, quantity does not fall.
- At T0: do nothing; sell forward at **USD 1.54/EUR**; or buy a put on USD.
- If unhedged at T0, at T1: remain unhedged or sell forward at **USD 1.65/EUR**.
- T2 scenarios: **USD 1.40/EUR** (strong dollar, favourable), or **USD 1.75/EUR** (weak dollar, unfavourable).
- Use the worked solution's **USD 1.5758/EUR** initial/strike quote consistently. Earlier setup rounds it to 1.57; do not mix that rounded input into the precise solution.

9.2 Build the option calculation before the comparison

- **Step 1 - Direct strike:** K = 1/1.5758 = **EUR 0.6345983/USD**. This is the protected euro selling price.
- **Step 2 - Total premium:** USD 0.02 per USD sold × USD 100 = **USD 2**.
- **Step 3 - Convert premium at T0:** 2/1.5758 = **EUR 1.2691966**. This exercise deducts it directly, without a financing adjustment.
- **Step 4 - Good scenario 1.40:** direct spot 1/1.40 = 0.7142857 EUR/USD exceeds strike. Do not exercise. Net = 100/1.40 - 1.2691966 = **EUR 70.1594**.
- **Step 5 - Bad scenario 1.75:** direct spot 1/1.75 = 0.5714286 is below strike. Exercise. Net = 100/1.5758 - 1.2691966 = **EUR 62.1906**.

Option net receipt = $100 \times \max(1/R_T, 1/1.5758) - 2/1.5758$

R_T is indirect USD/EUR spot. Take reciprocals before comparing selling prices; choose the higher EUR/USD price; convert USD 100; deduct the euro premium.

9.3 Compare all strategies on the same basis

Realised receipts in EUR	Strong dollar: 1.40	Weak dollar: 1.75
No hedge	100/1.40 = **71.4286**	100/1.75 = **57.1429**
T0 forward	100/1.54 = **64.9351**	**64.9351**
T1 forward	100/1.65 = **60.6061**	**60.6061**
Put, net of premium	**70.1594**	**62.1906**

- **No hedge:** best in the favourable scenario, worst in the unfavourable one. The wide spread is the risk hedging reduces.
- **T0 forward:** fixed receipt higher than the put floor, but no stronger-dollar upside.
- **Put:** keeps most favourable FX gain after paying for protection. Only its floor is guaranteed, not the favourable outcome.
- **T1 forward:** fixes less revenue because the available quote is worse for selling dollars. It still protects against later adverse moves; it cannot reverse a movement already experienced.
- Consistent T0 quote 1.5758 gives gross budget receipts **100/1.5758 = EUR 63.4598**. The source's standalone EUR 62.5 label is not that conversion. The table uses one coherent set of inputs.
- These are **receipts, not operating margins**: recap supplies no operating cost to subtract. Do not import EUR 70 costs from the separate over-hedging exercise.

9.4 Receipts, effective rates and budget are not interchangeable

- **Market spot** is a price before option cost. **Net effective rate** is final receipts per foreign unit after premium.
- Good scenario: EUR 70.1594/USD 100 = **EUR 0.701594/USD**. Reciprocal **USD 1.425326/EUR**, not market spot 1.40.
- Floor: EUR 62.1906/USD 100 = **EUR 0.621906/USD**; reciprocal **USD 1.607959/EUR**.
- Divide net home receipts by foreign notional first; invert only to obtain the indirect effective quote.

10. Choose a hedge for the firm's objective

10.1 Judge the distribution, not one favourable scenario

> **Hedging:** reduces variability of expected cash flows about the mean of their distribution. In the lecture, reducing that variance reduces risk.

- No hedge leaves cash flows moving with FX. A good scenario may be attractive; a bad one may disrupt financing or operating plans.
- A matched forward fixes the conversion price. Certainty comes with forgone favourable FX movements.
- A bought option protects downside while retaining upside. Premium lowers exporter net receipts or raises importer total cost.
- Compare the guaranteed result, possible upside, protection cost and ability to absorb a bad outcome, not just the biggest number in one column.

10.2 Management judgement matters

- Aidan's decision combines **risk tolerance, FX view and confidence in that view**. A forecast does not make risk disappear.
- A firm prioritising a known cash amount can favour forward. A firm paying for flexibility can favour option, accepting its lower net floor or higher cost ceiling.
- The recap's reasons to stay unhedged include competitors not hedging; FX treated as core business risk; enough liquidity or debt/equity access to absorb losses; or shareholder diversification where exposure creates no indirect costs for the firm.
- Those are reasons to assess the situation, not proof that no hedge is always best. The firm's cash flow still bears the risk.

10.3 Place FX risk within financial risk management

- The lecture names market, credit, liquidity, operational, foreign exchange, interest rate, business, reputational and commodity risks. They are connected.
- A2A links **liquidity risk** to financial planning/pre-funding; **derating risk** to asset rotation and equity/hybrid financing; **interest rate risk** to the fixed/variable debt balance; and **sustainability covenant risk** to a sustainable financial committee.

10.4 Keep price and quantity risk together

- Budget hedging protects an earlier price but forecast amount may change. Invoice hedging uses a clearer contract amount but the rate may already have worsened.
- Hedging 100% of a forecast does not eliminate every risk. An excess derivative can become an uncovered obligation.
- A defensible recommendation identifies the cash flow, protection sought, matched amount/date and remaining risk. "Derivatives reduce risk" alone is not an explanation.

11. Common mistakes & exam traps

- **Wrong conversion:** USD ÷ (USD/EUR) or USD × (EUR/USD) gives EUR. Multiplying by USD/EUR does not.
- **Wrong appreciation label:** rising USD/EUR is strong EUR, weak USD. Rising EUR/USD means strong USD.
- **Wrong instrument:** receivable = short forward/bought put; payable = long forward/bought call on foreign currency.
- **Standalone derivative mistaken for total outcome:** a forward loss may offset a favourable business movement. Do not double-count payoff alongside physical settlement.
- **Gross strike mistaken for net protection:** put floor = QK minus premium; call ceiling = QK plus premium.
- **Premium disappears when unused:** false. It is paid in both exercise and non-exercise branches.
- **Strike confused with strategy break-even:** Boeing exporter boundaries are 1.46 and 1.48; Boeing importer 1.80 and 1.73092. Exercise and net-outcome comparison are different questions.
- **Percentage confused with per-unit premium:** Aidan's 1.5% applies to spot-valued notional. Boeing's 0.02 and 0.018 are USD per GBP.
- **Money dates mixed:** use Aidan maturity premium 16,840.50 and Boeing importer 95,400. Follow other examples' supplied direct-premium convention.
- **Cases mixed:** Aidan uses EUR; Boeing uses USD. Boeing exporter receives GBP 10m; importer pays GBP 5m. Recap sells USD 100.
- **Quantity risk forgotten:** the forward fixes price on its notional, not the sales amount. Lower sales do not cancel delivery obligations.
- **EUR 50 all called derivative loss:** over-hedge loss is EUR 11.5385; EUR 50 is total budget-margin deterioration, including revenue shortfall.
- **Rounded inputs mixed:** keep one quote consistently and round only at the end, especially reciprocals and recap 1.5758.

12. Must know absolutely

The complete logic

• Identify home currency, foreign amount, receipt/payment and settlement date.
• Read quote units. Direct and indirect quotations are reciprocals.
• Receive foreign currency: short forward or bought put. Pay foreign currency: long forward or bought call.
• A matched forward fixes the home-currency amount.
• Put: higher selling price minus premium. Call: lower buying price plus premium.
• Strike decides exercise. Equality of net outcomes decides forward-option break-even.
• Put premiums into the right currency and apply the stated time convention.
• Judge underlying and derivative together. Excess hedging creates an uncovered speculative position.

Sources and calculation basis

- **Corporate Financial Risk Management: Managing Transaction Exposure**, Rachele Anconetani, Financial Management & Markets, AFC 2026/2027, slides 1-47. Slides govern concepts, contract terms and interpretation.
- Course workbooks: **Aidan revised 25-09-2025**, **Boeing**, **Recap Exercise**, used to check calculations against the lecture.
- Numerical outcomes are recalculated from stated inputs. Source rounding and carrying conventions are reconciled beside the relevant example.